

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING LABORATORY

HMLAL701

LABORATORY MANUAL

B. Tech. Computer Engineering

INSTITUTE

Fr. C. Rodrigues Institute of Technology

Vashi, Navi Mumbai 400703

DEPARTMENT

Computer Engineering

Honours Level Laboratory | Credits: 2

Course Overview

Course Code: HMLAL701

Course Title: Artificial Intelligence & Machine Learning Laboratory

Credits: 02 | Examination: Continuous Assessment Only (50 Marks)

Pre-requisite: Python Laboratory (CESBL301 / equivalent)

Course Objectives

- Apply data pre-processing, visualization, and supervised learning for structured data.
- Design and train deep neural networks using modern optimization and regularization.
- Implement advanced architectures (CNN, RNN, Autoencoder, GAN) for real-world problems.
- Foster self-learning through global guided projects and certification-based enrichment.

CO-PO Mapping

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
HMLAL701.1	3	3	3	3	3	3					
HMLAL701.2	3	3	3	3	3	3	2	3			
HMLAL701.3	3	3	3	3	3	3	2	3			
HMLAL701.4		3	3		3	3	3	3			3
Average	3	3	3	3	3	3	3	3			3

Assessment Structure

Component	Description	Marks
Task Evaluation	Practical tasks 1–8	20
Guided Project	Practical 9 & 10 Coursera / Kaggle	20
Regularity & Participation	Attendance & Engagement	10
TOTAL		50

Practical Index

No.	Title	CO LO	Bloom Level
1	Develop A Machine Learning Pipeline For Healthcare Data	CO1 LO1.1	Apply
2	Evaluate The Various Classification Model Performance on Fintech Dataset	CO1 LO1.2	Evaluate
3	Build A Simple Neural Network Using Perceptron and MLP Models	CO2 LO2.1	Create
4	Create A Deep Neural Network For Image Classification	CO2 LO2.2	Create
5	Evaluate The Impact Of Regularization Techniques in Deep Neural Networks	CO2 LO2.3	Evaluate
6	Develop CNN-Based Healthcare Image Classification	CO3 LO3.1	Create
7	Create RNN And LSTM Model to Perform For Time-Series Forecasting	CO3 LO3.2	Create
8	Develop An Autoencoder And GAN for Medical Image Processing	CO3 LO3.3	Create
9	Self Learning – Online Course1 - Industry-recognized guided project or Coursera capstone in ML/DL/AI.	CO4 LO4.1	Create
10	Self Learning – Online Course 2- Industry-recognized guided project or Coursera capstone in ML/DL/AI.	CO4 LO4.1	Create

Industry Perspective Domain Mapping

P#	Industry Domain	Key Technique	Representative Company
1	Hospital Analytics	Regression Pipeline	Practo, Apollo AI
2	Fraud Detection	Classification + ROC-AUC	Razorpay, Paytm
3	Customer Segmentation	Perceptron, MLP	Amazon, Netflix

4	OCR & Autonomous Systems	DNNMNIST / CIFAR-10	India Post, HDFC Bank
5	Reliable AI Systems	Dropout, Weight Decay	Waymo, Google Cloud
6	Medical Diagnostics	CNN + Transfer Learning	Qure.ai, Niramai
7	Algorithmic Trading	RNN, LSTM	Zerodha, Goldman Sachs
8	Medical Image Enhancement	Autoencoder, GAN	GE HealthCare, Siemens
9	Industry Certification	Self-directed Learning	Coursera, Kaggle, fast.ai

PRACTICAL NO. 1

Healthcare ML Pipeline Ethical Data Handling and Regression Modelling

Course Outcome	Learning Outcome	Performance Indicators
CO1: Create and evaluate end-to-end ML solutions with responsible data handling	LO 1.1: Create an ethical, privacy-preserving ML pipeline for healthcare data	PI-1.1.1, 1.4.1, 2.1.2, 2.4.1, 5.1.1, 5.2.2, 6.3.1, 7.1.1, 7.2.1, 8.2.1

1. Aim / Objective

To build an ethical, privacy-preserving machine-learning pipeline using a healthcare dataset; apply data loading, preprocessing (cleaning, imputation, scaling), exploratory visualization, feature engineering, and regression modeling while documenting societal and clinical implications.

2. Rationale

Healthcare is one of the most impactful domains where ML can save lives. Yet it is also one of the most sensitive patient data carries privacy obligations under regulations like HIPAA and India's DPDP Act 2023. This practical bridges the gap between theoretical ML knowledge from Semester VI and real-world, ethical implementation. Students learn not just to build a pipeline but to ask: Who owns this data? Can our model harm vulnerable patients? How do we document our choices?

Real-world relevance: Hospitals use regression models to predict patient readmission risk, ICU stay duration, and resource allocation all directly impacting patient outcomes and hospital costs.

3. SDG Mapping

SDG No.	SDG Title	Mapping Level	Justification
SDG 3	Good Health and Well-Being	High	Uses healthcare datasets for disease prediction, supporting better clinical decision-making and health outcomes.
SDG 10	Reduced Inequalities	Medium	Ethical, bias-aware modelling ensures healthcare ML tools serve all demographic groups fairly.
SDG 4	Quality Education	High	Focusing on ML pipeline.

4. Prerequisite Knowledge

- Python programming (loops, functions, OOP basics)
- Descriptive statistics: mean, median, standard deviation, correlation
- Supervised learning concepts: regression, train-test split, cross-validation
- Pandas, NumPy, Matplotlib, Seaborn fundamentals
- Awareness of healthcare data privacy (HIPAA, DPDP Act 2023)

5. Short Theory

5.1 The ML Pipeline

A Machine Learning pipeline is a structured sequence of steps that transforms raw data into a deployable model. For healthcare data, this pipeline must incorporate privacy-by-design principles at every stage.

5.2 Key Steps

- Data Collection & Loading: Source data from public repositories (Kaggle, UCI ML Repository). Anonymize PII fields.
- Exploratory Data Analysis (EDA): Understand distributions, identify outliers, visualize class imbalance.
- Data Preprocessing: Handle missing values (mean/median/KNN imputation), remove duplicates, fix data types.
- Feature Engineering: Create new features, encode categorical data, normalize/standardize numeric.
- Model Training: Apply Linear Regression to predict continuous targets (e.g., patient LOS).
- Model Evaluation: Use MAE, MSE, RMSE, R^2 as metrics.

5.3 Linear Regression Formula

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where $\hat{y}$ is the predicted output, β_0 is the intercept, and $\beta_1 \dots \beta_n$ are learned coefficients for features $x_1 \dots x_n$.

5.4 Evaluation Metrics

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i| \quad | \quad \text{RMSE} = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]} \quad | \quad R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2]$$

5.5 Privacy-Preserving Practices

- Remove direct identifiers (name, Aadhaar, phone) before analysis.
- Use k-anonymity: ensure each record is indistinguishable from at least k-1 others.
- Document data provenance and usage purpose.

6. Software and Tools Required

- Python 3.10+
- Google Colab (free cloud GPU/CPU)
- scikit-learn 1.4+
- Pandas, NumPy
- Matplotlib, Seaborn
- Kaggle API (for dataset download)

7. Dataset Description

Dataset: Diabetes Health Indicators Dataset (Kaggle / UCI)

Source: <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>

Size: ~70,000 records, 21 features

Target Variable: Diabetes_binary (for classification) or BMI (for regression task)

Key Features: HighBP, HighChol, CholCheck, BMI, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbcCost, GenHlth, MentHlth, PhysHlth, DiffWalk, Sex, Age, Education, Income

8. Problem Statement

A regional hospital network wants to predict the BMI of patients based on their lifestyle and health indicators to identify patients at risk of diabetes-related complications. Build an ethical ML regression pipeline that preprocesses the data responsibly, engineers meaningful features, trains a Linear Regression model, and delivers interpretable predictions with full documentation of societal implications.

9. Step-by-Step Procedure

1. Open Google Colab: go to colab.research.google.com and create a new notebook.
2. Mount Google Drive (optional) or upload dataset directly.
3. Install dependencies: `!pip install scikit-learn pandas matplotlib seaborn`
4. Import libraries: `pandas, numpy, sklearn, matplotlib.pyplot, seaborn`
5. Load dataset using `pd.read_csv()` and display shape and `head(10)`.
6. EDA: Plot histograms, correlation heatmap, and check for missing values using `df.isnull().sum()`.
7. Privacy audit: Drop or mask any PII columns. Document which columns were removed and why.
8. Handle missing values: use `SimpleImputer` with `strategy='mean'` for numerical columns.
9. Feature Engineering: Create `BMI_Category` (underweight/normal/overweight/obese) from BMI column.
10. Scale features: Apply `StandardScaler` to all numerical input features.
11. Split data: `train_test_split` with `test_size=0.2` and `random_state=42`.
12. Train Linear Regression model: `model.fit(X_train, y_train)`.
13. Predict on test set and compute MAE, MSE, RMSE, R^2 using `sklearn.metrics`.
14. Plot: (a) Actual vs Predicted scatter plot, (b) Residual plot, (c) Feature importance (coefficients).
15. Document ethical choices made during preprocessing in a Markdown cell.
16. Save model as `diabetes_regression.pkl` using `joblib.dump()`.

10. Algorithm / Workflow

ALGORITHMHealthcare Regression Pipeline:

1. INPUT: Raw healthcare dataset (CSV file)
2. PROCESS:
 - a. Load → EDA → Identify missing values and outliers
 - b. Privacy Audit → Remove PII → Document removals
 - c. Impute missing → Encode categoricals → Scale numerics
 - d. Feature Engineering → Correlation analysis → Feature selection
 - e. Train-Test Split (80:20)
 - f. Fit Linear Regression → Predict → Evaluate
3. OUTPUT: MAE, RMSE, R^2 scores + Visualizations + Ethical Documentation

11. Expected Output

Correlation heatmap showing feature relationships

Training and test set sizes printed

MAE, RMSE, R^2 values printed in console

Actual vs Predicted scatter plot (points near diagonal = good fit)

Residual distribution histogram (should be approximately normal, centered at 0)

Feature coefficient bar chart (shows direction and magnitude of each feature's influence)

12. Result Analysis (Should be written in conclusion)

Answer the following based on your experiment:

- What is the R^2 score of your model and what does it mean?
- Which features have the highest positive and negative coefficients? What does this imply clinically?
- What was your MAE? Is this acceptable in a healthcare context?
- What ethical concerns arose during pre processing? How did you address them?
- How would you improve the model's performance? (suggest 2 techniques)

13. Ethical, Societal and Environmental Reflection

Using ML pipelines for healthcare prediction raises serious concerns about patient privacy, consent, and algorithmic bias.

Privacy: Patient data must be anonymized. Even after removing names, combinations of age, gender, and location can re-identify individuals (re-identification risk).

Bias: If training data over represents a demographic (e.g., urban, male, adult patients), the model performs poorly for underrepresented groups, potentially worsening health disparities.

Fairness: Healthcare ML must not discriminate based on protected attributes. Audit models using fairness metrics (demographic parity, equal opportunity).

Environmental Impact: Training large ML models on cloud GPUs has a carbon footprint. Use efficient models and consider green cloud providers.

Accountability: Clinicians must understand and be able to override ML recommendations. Explainability (using SHAP or LIME) is essential.

PRACTICAL NO. 2

FinTech Classification Logistic Regression, SVM, and XGBoost with ROC-AUC Analysis

Course Outcome	Learning Outcome	Performance Indicators
CO1: Create and evaluate end-to-end ML solutions with responsible data handling	LO 1.2: Evaluate and compare classification models on FinTech data using confusion matrix and ROC-AUC	PI-1.1.1, 1.4.1, 2.1.2, 2.4.1, 5.1.1, 5.2.2, 6.3.1, 6.3.2, 7.1.1, 7.2.1, 8.2.1

1. Aim / Objective

To evaluate and compare Logistic Regression, Support Vector Machine (SVM), and XGBoost classifiers on a financial fraud detection dataset, using confusion-matrix metrics and ROC-AUC curves to interpret model effectiveness.

2. Rationale

Financial fraud costs the global economy over \$5 trillion annually. Banks and FinTech companies deploy ML classifiers to detect fraudulent transactions in real-time often with sub-millisecond latency requirements.

This practical builds on the ML Pipeline skills from Practical 1 and introduces multi-model comparison critical skill for data scientists who must justify model selection to business stakeholders.

Ethical significance: False negatives (missed fraud) cost the bank money; false positives (wrongly blocking genuine transactions) frustrate customers and may disproportionately impact low-income users.

3. SDG Mapping

SDG No.	SDG Title	Mapping Level	Justification
SDG 1	No Poverty	Medium	Fraud detection protects financially vulnerable individuals from scams and financial crime.
SDG 8	Decent Work and Economic Growth	High	Reliable FinTech systems support economic activity and trust in digital financial infrastructure.
SDG 16	Peace, Justice and Strong Institutions	High	Combating financial fraud strengthens rule of law and institutional integrity.

4. Prerequisite Knowledge

- Logistic Regression and sigmoid function
- SVM: hyperplane, support vectors, kernel trick (linear, RBF)
- Ensemble methods: boosting, XGBoost decision tree structure
- Confusion matrix: TP, TN, FP, FN; Precision, Recall, F1-Score
- ROC curve and AUC interpretation

- Class imbalance handling: SMOTE, class_weight parameter

5. Short Theory

5.1 Classification in FinTech

Classification assigns each transaction to a class: legitimate (0) or fraudulent (1). Financial datasets are typically highly imbalanced; fraud may be < 0.1% of transactions.

5.2 Logistic Regression

Models $P(y=1|X)$ using the sigmoid function: $\sigma(z) = 1 / (1 + e^{-z})$. Interpretable, fast, and effective as a baseline. Assumes linearly separable classes.

5.3 Support Vector Machine (SVM)

Finds the optimal hyperplane that maximizes the margin between classes. The RBF kernel maps data to higher dimensions, enabling non-linear separation. C parameter controls regularization.

5.4 XGBoost

Gradient boosted decision trees. Each tree corrects errors of the previous. Key parameters: n_estimators (number of trees), max_depth, learning_rate, scale_pos_weight (for class imbalance).

5.5 Evaluation Metrics

Precision = $TP / (TP + FP)$ | Recall = $TP / (TP + FN)$ | $F1 = 2 \times (P \times R) / (P + R)$

AUC-ROC: Area Under the ROC Curve. AUC = 1.0 → perfect; AUC = 0.5 → random classifier.

5.6 Class Imbalance

Use SMOTE (Synthetic Minority Oversampling Technique) or set class_weight='balanced' in sklearn to prevent models from ignoring the minority (fraud) class.

6. Software and Tools Required

- Python 3.10+
- Google Colab
- scikit-learn 1.4+
- XGBoost (pip install xgboost)
- imbalanced-learn (pip install imbalanced-learn)
- Pandas, NumPy, Matplotlib, Seaborn

7. Dataset Description

Dataset: Credit Card Fraud Detection Dataset (Kaggle)

Source: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Size: 284,807 transactions, 31 features (V1–V28 are PCA-transformed, Time, Amount, Class)

Target Variable: Class (0 = Legitimate, 1 = Fraud)

Class Imbalance: 99.83% legitimate, 0.17% fraud highly imbalanced

8. Problem Statement

A FinTech company processing millions of digital transactions daily wants to build a real-time fraud detection classifier. Compare Logistic Regression, SVM, and XGBoost models on the credit card fraud

dataset. Handle class imbalance appropriately, evaluate all models using confusion matrix metrics and ROC-AUC curves, and recommend the best model with justification.

9. Step-by-Step Procedure

1. Open Google Colab and install required libraries: xgboost, imbalanced-learn.
2. Upload the creditcard.csv file to Colab or load via Kaggle API.
3. EDA: Check shape, class distribution (df['Class'].value_counts()), and summary statistics.
4. Visualize class imbalance using a bar/pie chart.
5. Preprocessing: Normalize 'Amount' and 'Time' using StandardScaler. Drop irrelevant columns.
6. Handle class imbalance: Apply SMOTE to training data only (never on test data).
7. Split data: 80% train, 20% test (stratified split to preserve class ratio).
8. Train Logistic Regression: LogisticRegression(class_weight='balanced', max_iter=1000).
9. Train SVM: SVC(kernel='rbf', class_weight='balanced', probability=True).
10. Train XGBoost: XGBClassifier(scale_pos_weight=ratio, use_label_encoder=False).
11. For each model: generate confusion matrix and classification report.
12. For each model: plot ROC curve and compute AUC score.
13. Plot all three ROC curves on a single figure for comparison.
14. Tabulate all metrics: Precision, Recall, F1, AUC for all three models.
15. Write a Model Selection Justification paragraph (which model and why).

10. Algorithm / Workflow

ALGORITHM FinTech Classification Pipeline:

1. INPUT: Raw credit card transaction data (CSV)
2. PROCESS:
 - a. EDA → Visualize class imbalance
 - b. Normalize Amount, Time → Train-test split (stratified)
 - c. Apply SMOTE on training set only
 - d. Train: LR → SVM → XGBoost
 - e. For each model: Predict → Confusion Matrix → Classification Report → ROC Curve
3. OUTPUT: Comparative metrics table + Multi-ROC plot + Model recommendation

11. Expected Output

Class distribution bar chart (showing severe imbalance)

After-SMOTE class distribution (balanced)

Confusion matrix for each of the 3 models

ROC curves for all 3 models on a single plot

Comparison table: Precision, Recall, F1, AUC for all models

12. Result Analysis (to be written as conclusion)

Answer the following based on your experiment:

- Which model achieved the highest AUC? What does AUC measure?
- Compare Recall scores across models. Why is high Recall critical in fraud detection?
- What happens if you do NOT apply SMOTE? Run the experiment and compare.

- Which model would you deploy in production and why? Consider both performance and speed.
- What are the trade-offs between Precision and Recall in this fraud detection context?

13. Ethical, Societal and Environmental Reflection

FinTech ML raises profound ethical concerns that go beyond technical accuracy.

False Positives in fraud detection: Blocking a legitimate transaction can leave a customer stranded especially dangerous for low-income users who have no backup payment method. This is a justice and equity issue.

Algorithmic Bias: If training data reflects historical discrimination (e.g., certain ethnicities flagged more often), the model will perpetuate this bias. Models must be audited for protected attribute fairness.

Explainability: Customers whose cards are blocked deserve a reason. The EU AI Act (2024) mandates explainability for high-risk AI systems including credit/fraud scoring.

Data Sovereignty: Transaction data is highly sensitive. Who stores it? In which country? Under what regulations? India's DPDP Act 2023 restricts cross-border data transfer.

Environmental Cost: XGBoost with thousands of trees has higher energy consumption than Logistic Regression. For real-time deployment at scale, energy efficiency is a legitimate design consideration.